

Bayesian leak tightness assessment of nuclear containment buildings

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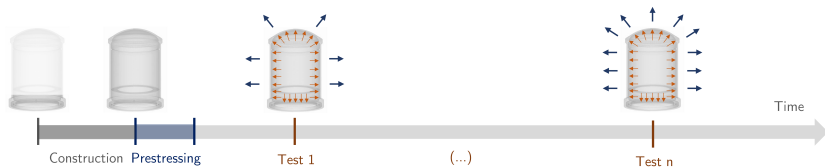
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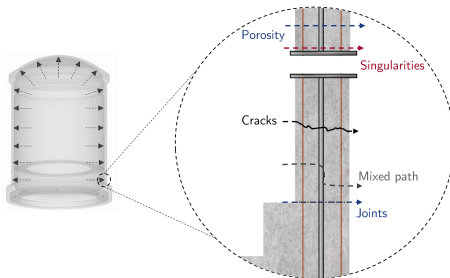
Introduction

- Double-walled **Nuclear Containment Buildings (NCB)** are reinforced and prestressed concrete structures...
- ...whose leak tightness plays a key role in the safety of their surrounding environment
- Leak tightness of NCB is assessed through periodic **pressurization tests** aiming at measuring their leakage rate



Introduction

- Leak tightness of NCB may evolve in time, under the action of several physical processes related to **concrete ageing**



- Necessity to develop a **probabilistic forecasting methodology** aiming at assessing the **risks** of exceeding a critical leakage rate threshold...
- ...whose predictions may be **updated** by assimilating **monitoring data** (delayed strains, leakage rate)

Forward model, prior distribution and likelihood function

- Let $\mathcal{M} : \mathcal{D}_{\mathbf{x}} \longrightarrow \mathbb{R}^n$ be a **forward model** which maps a set of **input parameters** $\mathbf{x} \in \mathcal{D}_{\mathbf{x}} \subset \mathbb{R}^d$ to a **response** $\mathcal{M}(\mathbf{x}) \in \mathbb{R}^n$

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 $\Rightarrow$ definition of a **likelihood function** $\mathcal{L}_y(x)$

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- Additive Gaussian model widespread used:

$$y = \mathcal{M}(x) + e \quad \text{with } e \sim \mathcal{N}(\mathbf{0}, \Sigma_e)$$

$$\Rightarrow \mathcal{L}_y(x) = \det(2\pi\Sigma_e)^{-1} \exp\left(-\frac{1}{2}(\mathbf{y} - \mathcal{M}(x))^T \Sigma_e^{-1}(\mathbf{y} - \mathcal{M}(x))\right)$$

Posterior distribution and sampling methods

- ▶ Bayes' theorem $\Rightarrow$ **posterior density** of $\mathbf{X}$ knowing $\mathbf{Y} = \mathbf{y}$:

(Stuart, 2010)

$$\pi(\mathbf{x}|\mathbf{y}) = \frac{\pi_0(\mathbf{x})\mathcal{L}_{\mathbf{y}}(\mathbf{x})}{Z_{\mathbf{y}}}$$

where $Z_{\mathbf{y}}$ is the **model evidence**:

$$Z_{\mathbf{y}} = \int_{\mathcal{D}_{\mathbf{x}}} \mathcal{L}_{\mathbf{y}}(\mathbf{x})\pi_0(\mathbf{x})d\mathbf{x}$$

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(Robert & Casella, 2004; Brooks et al., 2011)

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- **Transitional MCMC (TMCMC)**: recent efficient algorithm for sampling from posteriors with complex shapes (e.g. with multi-modal or peaked densities)

(Ching & Chen, 2007; Betz et al., 2016)

Surrogate modeling for accelerated Bayesian computations

- (T-)MCMC sampling algorithms require a large number of model evaluations ($\sim 10^5 - 10^6$) $\Rightarrow$ acceleration by using **surrogate models**

(Marzouk et al., 2007; Marzouk & Najm, 2009; Angelikopoulos et al., 2015)

- A **surrogate model** $\hat{\mathcal{M}}$ is an approximation of the model $\mathcal{M}$, well-defined under some regularity and integrability conditions on $\mathcal{M}$
- Use of **Polynomial Chaos Expansions (PCE)**: (Ghanem & Spanos, 1991; Xiu & Karniadakis, 2002)

$$\hat{\mathcal{M}} = \sum_{j \in \mathcal{J}} c_j \Psi_j$$

- ▶ Ψ_j : **basis functions** (orthonormal polynomials, w.r.t. the **prior** distribution)
- ▶ c_j : **coefficients** (to be computed)
- ▶ $\mathcal{J} \subset \mathbb{N}^d$: set of **multi-indices**
- c_α computed by using sparse regression techniques, from a limited sample $\mathcal{X} = (\mathbf{x}^{(i)})_{i \in \llbracket 1, n \rrbracket}$ of input parameters $\Rightarrow$ **sparse PCE**

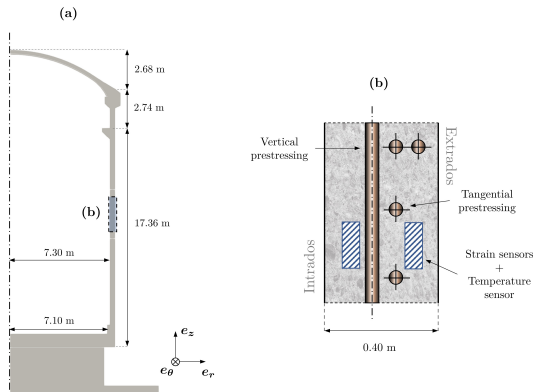
(Efron et al., 2004; Blatman & Sudret, 2011; Lüthen et al., 2021)

Application to a 1:3 scale NCB: the VeRCoRs mock-up

VeRCoRs mock-up: non-nuclear 1:3 scale NCB built by EDF company (Charpin et al., 2021)

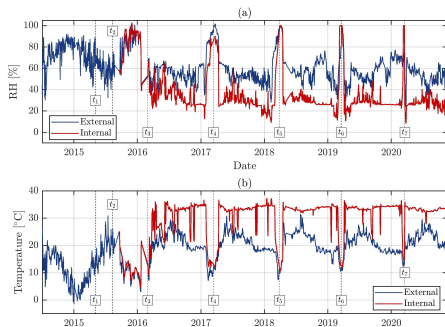


VeRCoRs mock-up ©RILEM

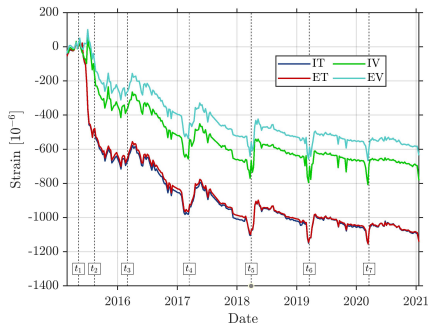


Application to a 1:3 scale NCB: the VeRCoRs mock-up

Well instrumented structure with a large amount of **monitoring data** (e.g. temperature, RH, strains, leakage)



Ambient air (a) RH (b) temperature measurements



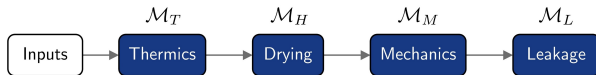
Mean delayed strains measurements of VeRCoRs standard zone.
Notations: I: intrados; E: extrados; T: tangential; V: vertical

THML forward model

- Use of a **chained** thermo-hydro-mechanical & leakage (THML) model:

(Bouhjiti et al., 2018; Rossat, Bouhjiti, et al., 2021)

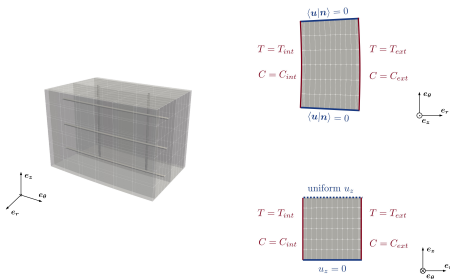
$$\mathcal{M}(\mathbf{x}) = (\mathcal{M}_L \circ \mathcal{M}_M \circ \mathcal{M}_H \circ \mathcal{M}_T)(\mathbf{x})$$



- **Thermal calculations:** linear heat equation
- **Drying calculations:** nonlinear diffusion equation
- **Mechanical calculations:** shrinkage, basic and desiccation creep models
- **Leakage calculations:** nonlinear diffusion equation for diffuse leaks, phenomenological model for local leaks (Rossat, Baroth, Briffaut, Dufour, Monteil, et al., 2021)

Finite element model and input parameters

- Use of **Representative Structural Volumes (RSV)** for modeling the behavior of the mock-up
(Jason et al., 2007; Rossat, Baroth, Briffaut, Dufour, Monteil, et al., 2021)



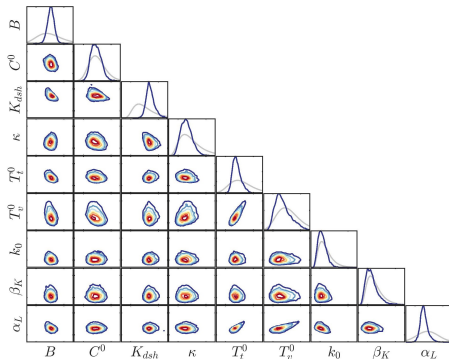
Finite element mesh of VeRCoRs standard zone RSV & boundary conditions

- Identification of 9 important input parameters:
 - ▶ 2 drying parameters: (B, C^0)
 - ▶ 4 mechanical parameters: $(K_{dsh}, \kappa, T_t^0, T_v^0)$
 - ▶ 3 leakage parameters: (k_0, β_K, α_L)

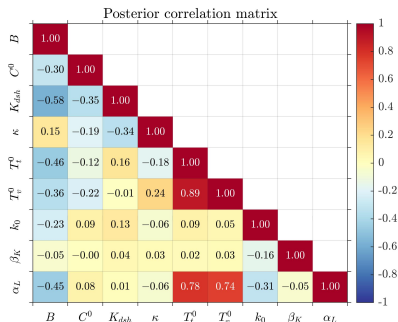
Posterior input parameters

- Bayesian updating of model parameters using strain and leakage monitoring data

Posterior - Step 1



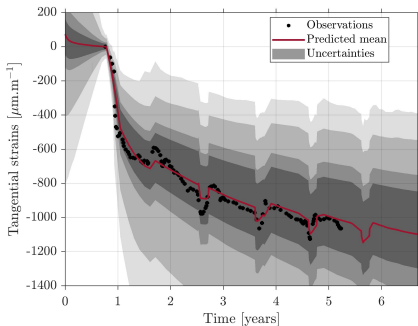
Posterior univariate and bivariate marginal densities (*Kernel Density Estimations constructed from TCMC samples*)



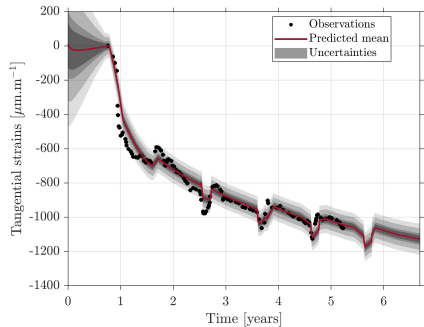
Posterior correlation matrix of model parameters

Probabilistic delayed strains predictions

■ Probabilistic predictions of tangential delayed strains



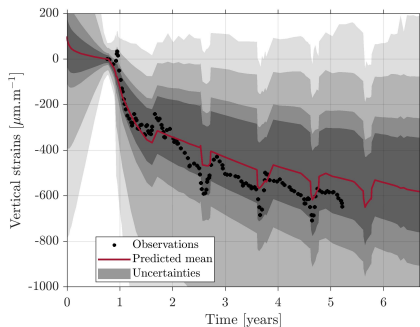
Prior tangential strains (Mean & 50, 70, 95, 99.5% quantiles)



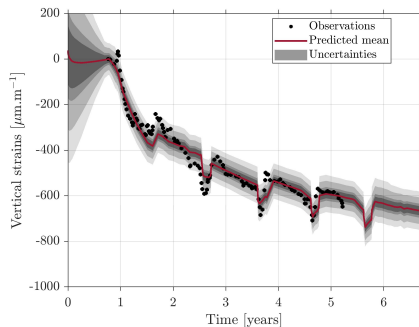
Posterior tangential strains (Mean & 50, 70, 95, 99.5% quantiles)

Probabilistic delayed strains predictions

■ Probabilistic predictions of vertical delayed strains



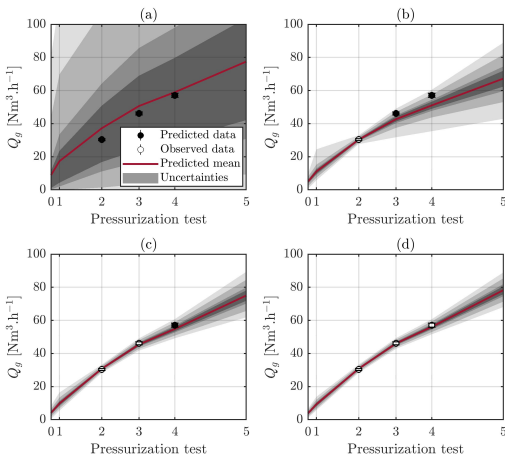
Prior vertical strains (Mean & 50, 70, 95, 99.5% quantiles)



Posterior vertical strains (Mean & 50, 70, 95, 99.5% quantiles)

Probabilistic global leakage predictions

- Probabilistic predictions of the **global leakage rate** of VeRCoRs mock-up
- Bayesian updating of predictions after each pressurization test



Conclusions

- Methodology for probabilistic predictions of the leakage behavior of NCB with *in-situ* monitoring data assimilation in a Bayesian framework
- Combination of sparse PCE and a TMCMC sampler $\Rightarrow$ affordable computational cost
 - ▶ PCE construction: $\sim 3h$ by performing parallel computations on 8 CPUs
 - ▶ TMCMC sampling: $\sim 1.5h$ when replacing the FE model by sparse PCE
- Perspectives:
 - ▶ the presented methodology is based on **prior-based** PCE
 - ▶ PCE-accelerated Bayesian inversions may fail if the posterior mass is located in zones where PCE surrogates are inaccurate
 - ▶ Development of a Bayesian updating methodology based on **adaptive surrogate models**

(Rossat, Baroth, Briffaut, & Dufour, 2021)

Thank you for your attention !

Likelihood function

- Observations $\mathbf{Y}$ of the system response may be linked to input parameters $\mathbf{X}$ by modeling the **conditional distribution** of $\mathbf{Y}$ knowing $\mathbf{X}$

⇒ Additive Gaussian model usually considered:

$$\mathbf{y} = \mathcal{M}(\mathbf{x}) + \mathbf{e}$$

$\mathbf{y}$: observation data

$\mathbf{e} \sim \mathcal{N}(\mathbf{0}, \Sigma_e)$: discrepancy term

Σ_e : discrepancy covariance matrix

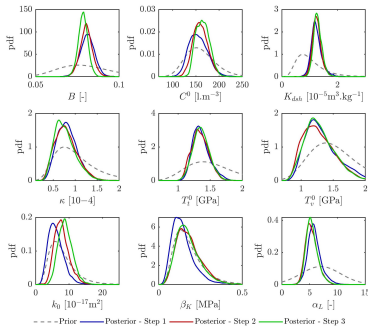
- ▶ The **likelihood function** is given by the conditional PDF of $\mathbf{Y}|\mathbf{X}$:

$$\mathcal{L}_{\mathbf{y}}(\mathbf{x}) = \mathcal{N}(\mathbf{y}|\mathcal{M}(\mathbf{x}), \Sigma_e)$$

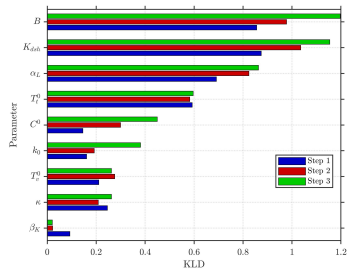
Posterior input parameters

- Bayesian updating in 3 steps, from Test 3 to Test 5
- Quantification of the information gain by using **Kullback-Leibler Divergence (KLD)** (or **relative entropy**):
(Kullback & Leibler, 1951)

$$\mathcal{D}_{KL}(\alpha||\beta) = \int \alpha(\mathbf{x}) \log \left(\frac{\alpha(\mathbf{x})}{\beta(\mathbf{x})} \right) d\mathbf{x}$$



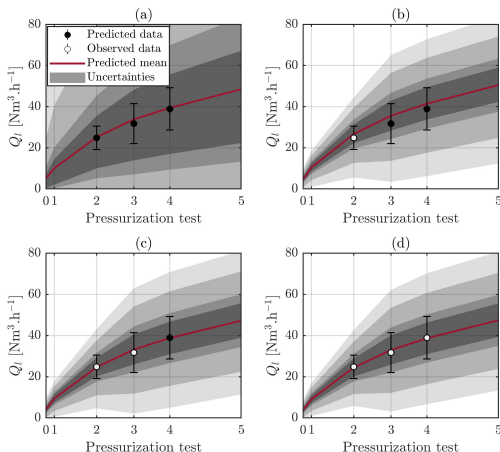
Posterior univariate marginal densities (Kernel Density Estimations constructed from TMCMC samples)



Kullback-Leibler Divergence of posterior 1D marginals w.r.t. their prior counterpart

Probabilistic local leakage predictions

- Probabilistic predictions of the **cumulated local leaks** of VeRCoRs mock-up
- Bayesian updating of predictions after each pressurization test



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